Exploring the use of LID in learning disentangled and cluster-friendly latent spaces in Autoencoders

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Course: Dimensionality and Scalability in AI (CS786854)
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Problem Statement, Significance and Goals

Problem Statement:

- To explore the various ways we can utilize local intrinsic dimensionality (LID) of data in latent space to achieve disentangled or cluster-friendly latent spaces

Needs and Significance:

- Latent representations encoded by deep neural networks are often an entangled mesh of variations coming from different features
- Dimension reduction is a crucial preprocessing step in clustering problems
- The latent space produced by such reductions, may not be cluster friendly though

Project Goals:

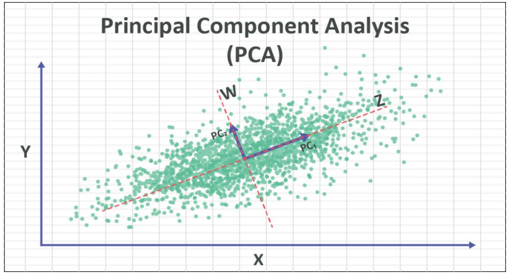
- First investigating both theoretically and empirically the prospect of learning disentangled representations by utilizing LID via Total Correlation. Then investigating how LID can help form smoother, more separable clusters in the latent space produced by an Autoencoder.

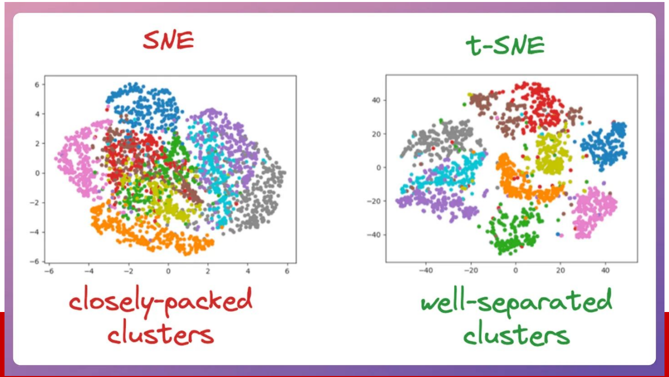
Introduction

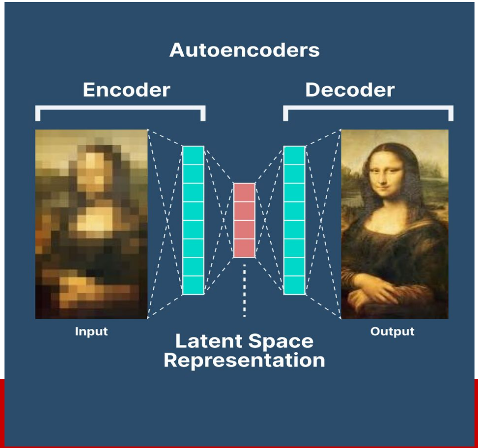
Understanding Dimension Reduction

- Most datasets (e.g., images or sensor data) are in high-dimensional spaces but lie on a lower-dimensional manifold.
- The curse that plagues data scientists - “Curse of dimensionality”
- Lot of noisy and redundant features in higher dimensions
- Problem of visualization and interpretation

Existing models for Dimension Reduction include:





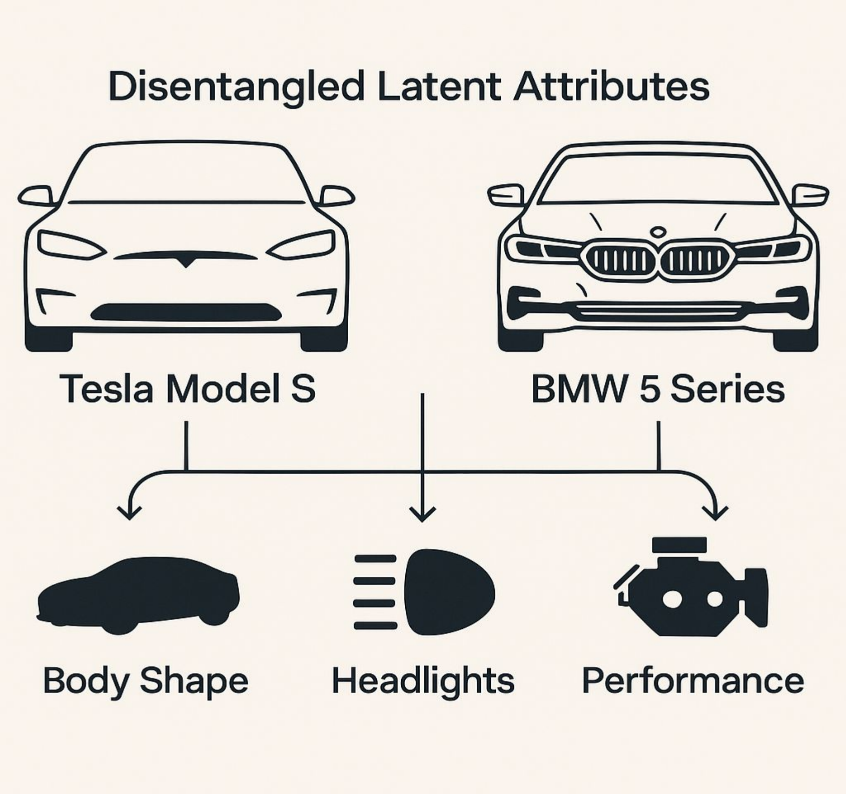


Introduction

The problem of entanglement

- Autoencoders are trained to compress input data into such latent representations, retaining only the most essential features while filtering out extraneous or redundant dimensions.
- The representations are still an entangled ensemble of variations coming from various features of the data
- Makes it harder to directly interpret the representations
- Thus it is essential to enforce disentanglement in order to increase the interpretability

What is disentanglement?

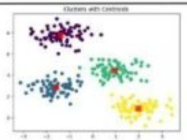
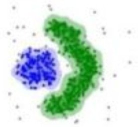
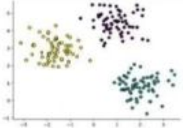
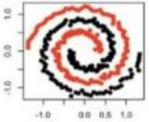


Introduction

Clustering in latent spaces

- Clustering is an unsupervised machine learning technique used to group similar unlabeled data points into clusters based on defined similarity measures.
- Clustering high dimensional data poses several challenges
- The learnt representations are not inherently cluster friendly
- They aren’t guaranteed to align with meaningful or well-separated clusters unless explicitly guided to do so

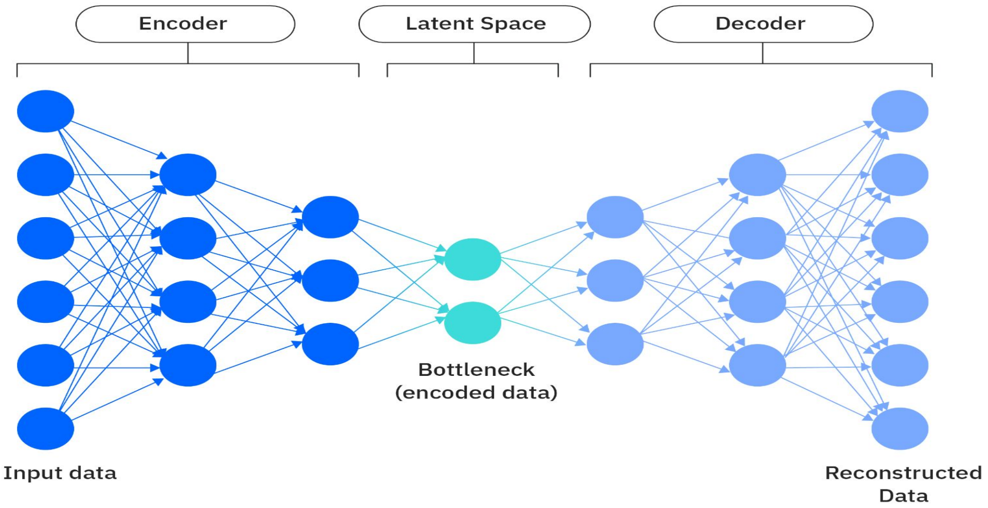
Types of clustering

Representation	Algorithm Name	Hyperparameter
	K-Means Clustering	Partitions data into K clusters by minimizing variance within each cluster.
	Hierarchical Clustering	Builds a hierarchy of clusters by iteratively merging or splitting existing groups.
	DBSCAN	Forms clusters based on density; groups densely packed points and marks outliers.
	Mean Shift	Finds clusters by locating and adapting to the centroids of data points.
	Spectral Clustering	Uses eigenvalues of similarity matrix to reduce dimensions before clustering.

Background

Autoencoders

- Autoencoders are neural networks based dimensionality reduction models that capture nonlinear relationships in the data



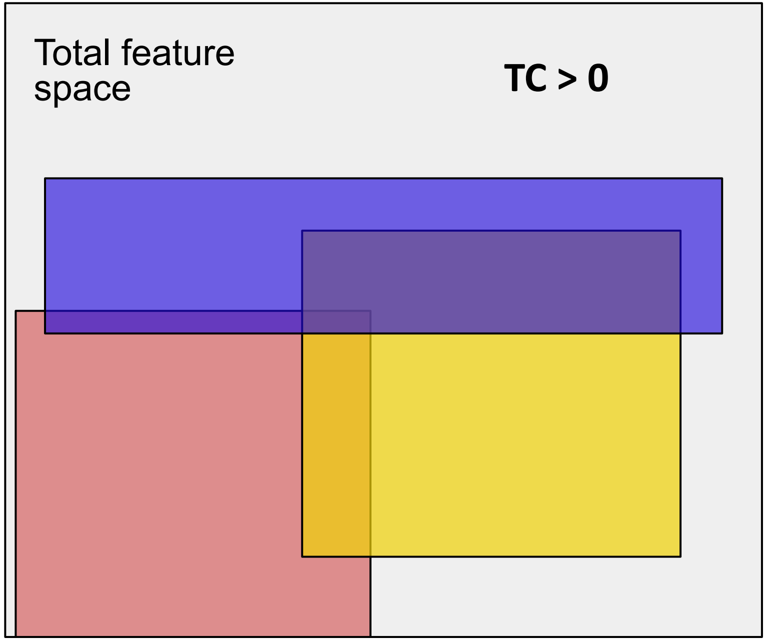
- An encoder responsible for compressing the data to latent space
- A bottleneck dimension characterizing the latent space
- A decoder for reconstructing the data from latent space
- Training involves minimizing the reconstruction error.

Background

Total Correlation

- Total Correlation (TC) quantifies the amount of total information in a set of variables beyond what would be there if the variables were independent.
- TC uses entropy as a measure of information contained in a variable.

$$TC(X_1, X_2, ..X_n) = \sum_{i=1}^{n} H(X_i) - H(X_1, X_2, ..X_n)$$

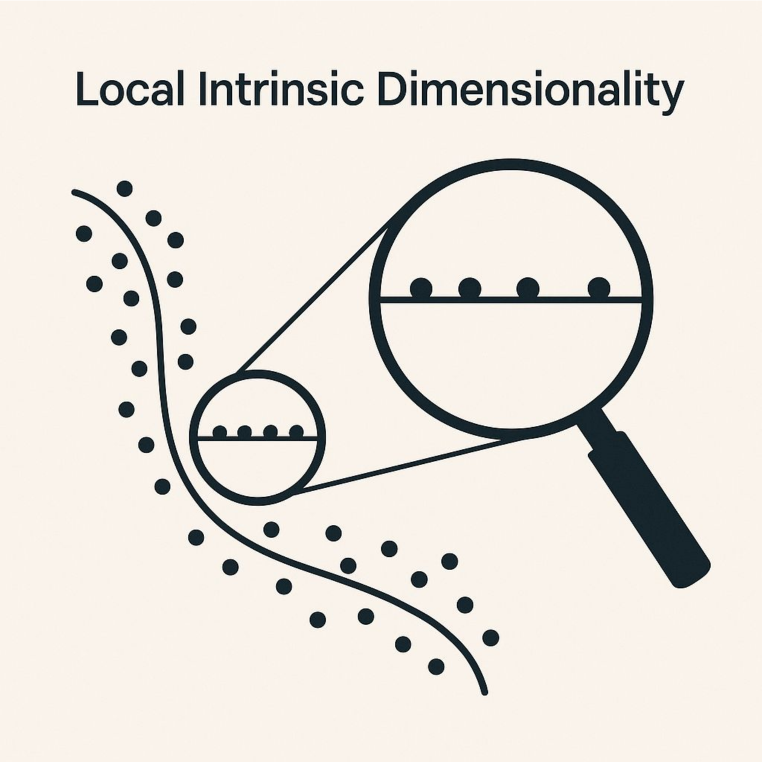


TC of this feature space is some positive value > 0. In a disentangled feature space, the TC measured will be 0

Background

Local Intrinsic Dimensionality (LID)

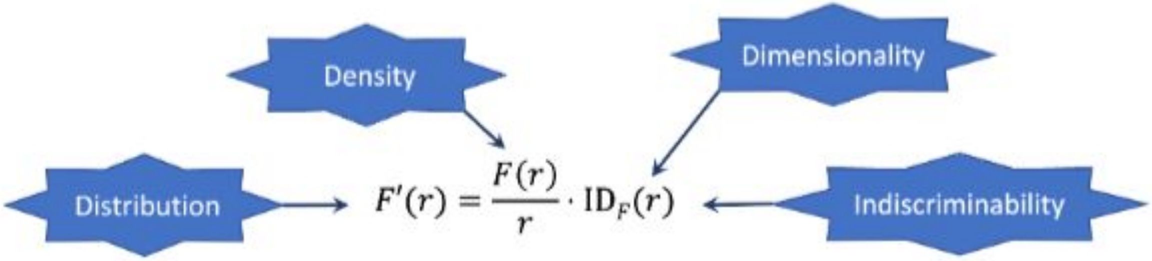
- Intrinsic dimensionality is the minimum number of independent variables required to describe a dataset without loss of its intrinsic structure.
- The winding curve on the left represents a high-dimensional data space
- Zooming in the manifold in which the data lies looks almost flat and one-dimensional in a tiny neighborhood. LID asks: *how many directions do I actually need, locally, to move around this point?*



Background

Local Intrinsic Dimensionality

- Mathematically, LID is the rate at which the neighborhood distribution around a point normalized by the density around the point at a specific distance r, given by:



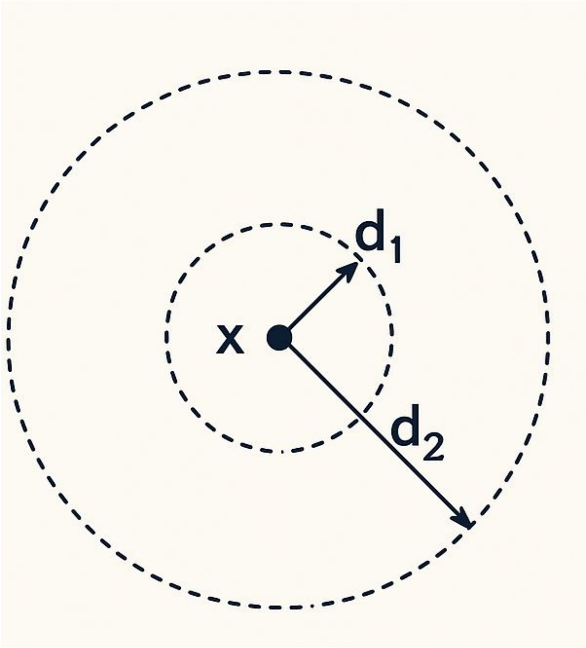
- As we tend r towards 0, we get the asymptotic tail distribution. It tells us in how many orthogonal directions we can move around a point. If we can move in just one direction (say up or down), the asymptotic LID around that point is 1.

Background

Two-NN estimator of LID

- There are various LID estimators such as Maximum Likelihood Estimator, Method of Moments Estimator, Two-NN Estimator.
- The Two-NN (Two Nearest Neighbor) estimator measures how quickly the volume (number of points) expands as we move away from a point *x*. With only two distances it supplies a surprisingly good maximum-likelihood estimate under mild assumptions, making it fast to compute even in high-dimensional spaces or large datasets

Two-NN estimator



Background

Weighted K-Means Clustering

- Unlike regular K-means, where cluster centers and cluster assignments are updated just on the basis of distance
- Weighted k-means assigns weights to points to define their probability of belonging to one cluster over the other. This influences the cluster center updates as follows:

$$v_r = \frac{\sum_{i=1}^{n} w_{ir} Z_i}{\sum_{i=1}^{n} w_{ir}}.$$

- Idea is to reduce the loss incurred by misassignment of points (now weighted as well)

$$L_{km} = \frac{1}{nv} \sum_{i=1}^{n} \sum_{r=1}^{v} w_{ir} ||z_i - c_r||^2$$

Different-sized dots show points with different weights

Methodology

Disentanglement using LID via Total Correlation

- Following the paper - *L. Basile, S. Acevedo, L. Bortolussi, F. Anselmi, and A. Rodriguez, "Intrinsic dimension correlation: uncovering nonlinear connections in multimodal representations," 2025* - it was intuitive to model Total Correlation using LID as a proxy for entropy as follows:

$$TC_{I_d}(X_1, X_2, ..X_n) = \sum_{i=1}^{n} I_d(X_i) - I_d(X_1, X_2, ..X_n)$$

- The correctness of this naive formulation is subject to question
- TC is defined over the full data distribution, whereas LID characterizes behavior in the lower distributional tail.
- For theoretical investigation, we have to study the behaviour of TC in the limit where the neighborhood radius approaches zero

Methodology

Disentanglement using LID via Total Correlation

$$\lim_{w->0} TC(Z_1, Z_2, ..Z_n; w) = \lim_{w->0} \sum_{i=1}^{n} H(Z_i; w) - H(Z_1, Z_2, ..Z_n; w)$$

$$= \lim_{w->0} \sum_{i-1}^{n} \left[-\int_0^w Z'_{iw}(t) \ln(Z'_{iw}(t)) \right] - \left[-\int_0^w Z'_w(t) \ln(Z'_w(t)) \right]$$

$$= \lim_{w->0} \sum_{i-1}^{n} \left[-\int_0^w ID_{Z_i}(t) \frac{Z_i(t)}{t} \ln\left(ID_{Z_i}(t) \frac{Z_i(t)}{t}\right) dt \right] - \left[-\int_0^w ID_Z(t) \frac{Z(t)}{t} \ln\left(ID_Z(t) \frac{Z(t)}{t}\right) dt \right]$$

$$= \lim_{w->0} \sum_{i-1}^{n} \left[-\int_0^w \frac{ID^*_{Z_i}}{t} \left(\frac{t}{w}\right)^{ID^*_{Z_i}} \ln\left(\frac{ID^*_{Z_i}}{t} \left(\frac{t}{w}\right)^{ID^*_{Z_i}} \right) dt \right] - \left[-\int_0^w \frac{ID^*_Z}{t} \left(\frac{t}{w}\right)^{ID^*_Z} \ln\left(\frac{ID^*_Z}{t} \left(\frac{t}{w}\right)^{ID^*_Z} \right) dt \right]$$

$$= \lim_{w->0} \sum_{i-1}^{n} \left[1 - \frac{1}{ID^*_{Z_i}} - \ln \frac{ID^*_{Z_i}}{w} \right] - \left[1 - \frac{1}{ID^*_Z} - \ln \frac{ID^*_Z}{w} \right]$$

TC diverges as neighborhood radius tends to 0

Methodology

Learning cluster-friendly latent space

- To ensure that the autoencoder learns a latent space that facilitates clustering, several modifications were introduced to the baseline architecture.
- During training, weighted K-means clustering was applied directly to the latent space whose loss gave us the first additional loss - the K-means loss:

$$L_{km} = \frac{1}{nv}\sum_{i=1}^{n}\sum_{r=1}^{v} w_{ir}||z_i - c_r||^2$$

- The weights are assigned using the softmax function such that it gives us the similarity measure between a point and its corresponding centroid, therefore the closer the point the higher is the weight:

$$w_{ir} = \frac{\exp\left(-||z_i - c_r||^2\right)}{\sum \exp\left(-||z_i - c_r||^2\right)}$$

Methodology

Learning cluster-friendly latent space

- To make sure similar points are clustered together and dissimilar points are as far apart as possible, a KL divergence loss function is introduced in models like t-SNE, scDeepCluster and scziDesk
- First a probability distribution of every point z_i belonging to the same cluster as other points in latent space is built using t-distribution kernel as follows:

$$p_{j|i} = \frac{(1 + ||z_i - z_j||^2/t)^{-(t+1)/2}}{\sum_{i \neq k}(1 + ||z_i - z_k||^2/t)^{-(t+1)/2}}$$

- An auxiliary target distribution is defined as follows:

$$q_{j|i} = \frac{p_{j|i}^2 \sum_{i \neq j} p_{j|i}}{\sum_{i \neq k} p_{k|i}^2 \sum_{i \neq j} p_{k|i}}$$

- The Kullback-Leibler (KL) divergence loss is employed to encourage the model distribution p to approximate the target distribution q.:

$$Loss_{KL} = \sum_{i=1}^{N} q_{j|i} \log \frac{q_{j|i}}{p_{j|i}}$$

Methodology

Learning cluster-friendly latent space

- Finally, the Itakura-Saito divergence using estimated LIDs of distributions $q_{j|i}$ and $p_{j|i}$ as is introduced as follows:

$$Loss_{IS} = \left(\frac{\widehat{ID^*_P}}{\widehat{ID^*_Q}}\right) - \log\left(\frac{\widehat{ID^*_P}}{\widehat{ID^*_Q}}\right) - 1$$

- This loss gives manifold awareness to the clusters - pulling the clusters toward a low (or uniform) target encourages the encoder to flatten out wrinkle-like variations inside a cluster, giving tighter, more linear latent shapes which are more separable.
- It counter-balances KL loss terms behavior of over-tightening dense regions and neglecting sparse ones by pushing sparse zones to contract as well
- It makes sure outliers (which generally have very high LID) from pulling the centroids towards them
- Low LID makes sure data is tightly packed - less room for adversaries

Methodology

Learning cluster-friendly latent space

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- Low LID makes sure data is tightly packed - less room for adversaries

Methodology

Two-NN estimator

- We can estimate Ids using Two-NN estimator as was discussed in Background.
- Briefly, the estimator operates by first computing the ratio between the distances to the second and first nearest neighbors of each data point, which is then transformed using an inverse logarithmic scale.
- Subsequently, it evaluates the cumulative distribution function (CDF) of these values
- relative to each point.
- A line is then fitted through the origin, using the inverse log-scale ratios as the x-coordinates and the corresponding CDF values as the y-coordinates.
- The slope of this line provides the estimated LID of the data. The use of only the two nearest neighbors renders the estimator highly efficient and free of tunable parameters.

Results

Experimental setup and Datasets

- A simple autoencoder architecture was developed, consisting of an encoder and decoder each containing a single hidden layer, alongside a central bottleneck layer.
- Network weights were optimized using the Adam optimizer.

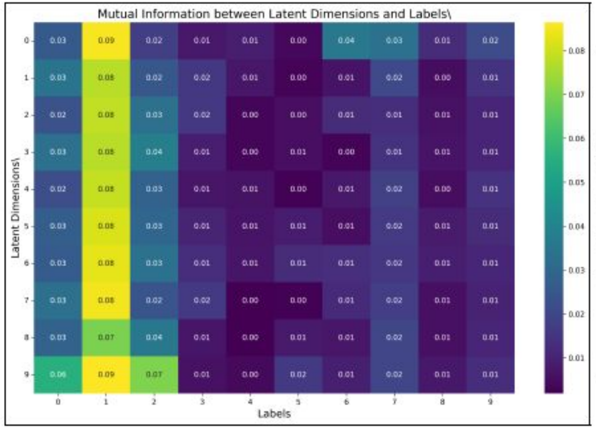
Dataset used	Samples	Features
MNIST	60000	784
Ship Performance	2736	24
Diabetes	152	20
Cover Type	581012	54
Market	2240	29
Fashion-MNIST	60000	784

TABLE I: Datasets

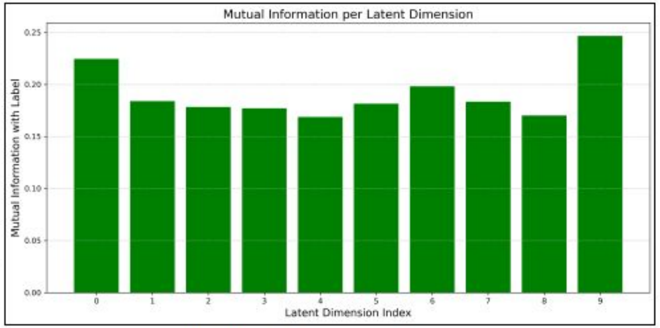
- Training was conducted utilizing an NVIDIA A100 GPU; however CPU-based training is also feasible, with the exception of one PyTorch library dependency within the LID estimator module, which explicitly requires CUDA support.
- The source code and the dataset are all available on GitHub

Results

Disentanglement using LID via Total Correlation



(a) Mutual Information Heatmap



(b) Mutual Information Scores

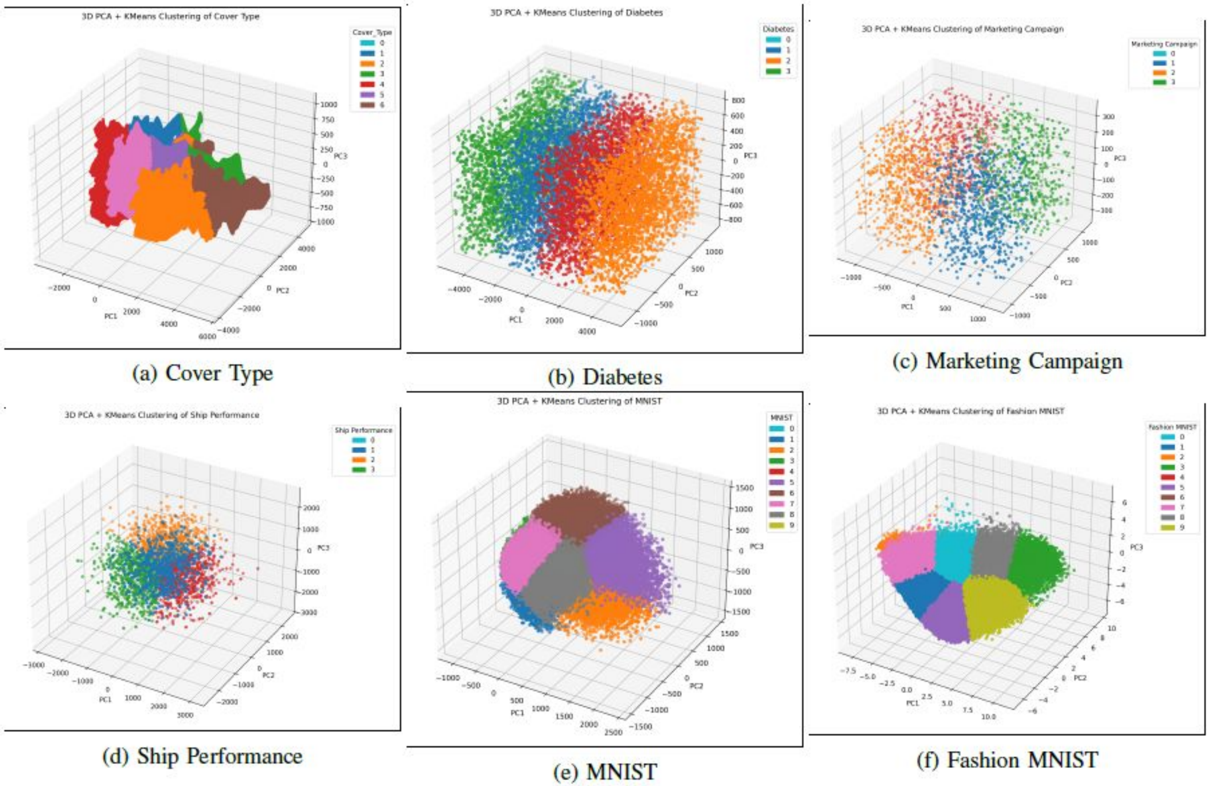
Fig. 1: The amount of disentanglement measured by (a) how a informative a particular latent dimension is in expressing a particular label and (b) how much mutual information is shared between a particular latent dimension and label '0'

- MNIST dataset was used for this experiment. In an ideally disentangled representation, each latent dimension would independently capture information pertinent to a single digit class.

Results

Learning cluster-friendly latent space

To establish a baseline, Principal Component Analysis (PCA) was performed to reduce each dataset to three dimensions, followed by K-means clustering.



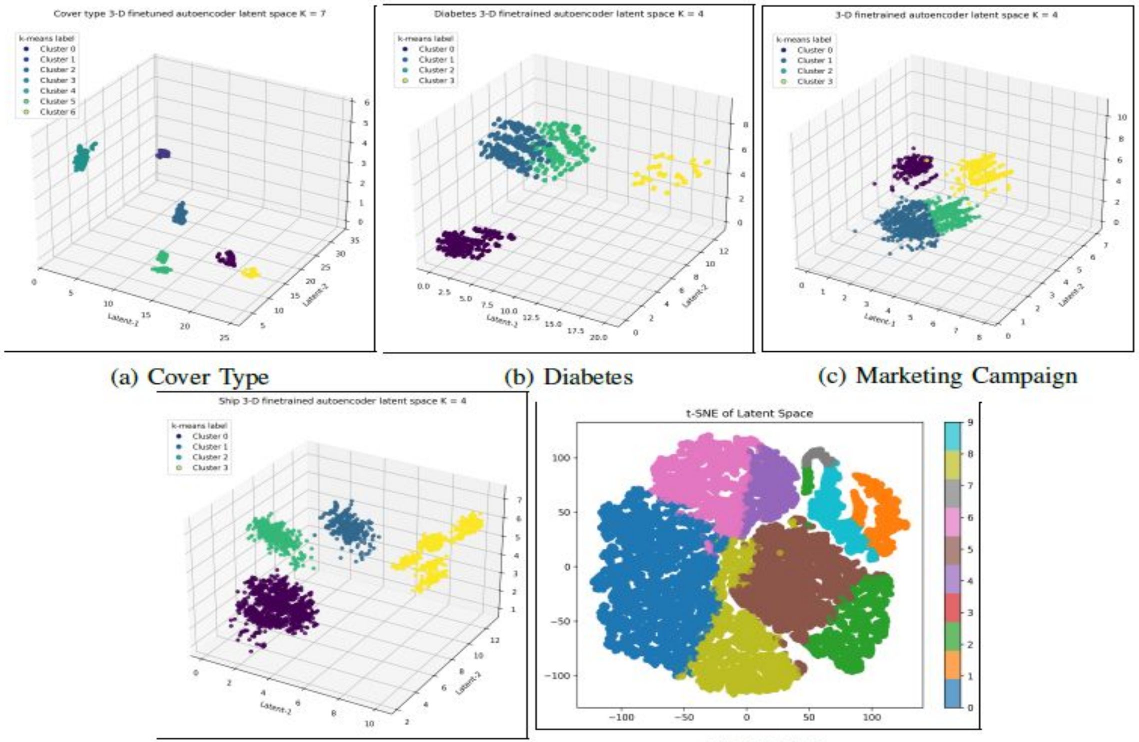
(a) Cover Type (b) Diabetes (c) Marketing Campaign (d) Ship Performance (e) MNIST (f) Fashion MNIST

Fig. 2: PCA + Kmeans performed as a baseline

Results

Learning cluster-friendly latent space

- These results are produced by a model that only uses the IS loss along with reconstruction and K-means losses.
- Strong performance on Cover Type and Ship Performance dataset.
- Satisfactory for Diabetes and Market.
- Better than baseline for MNIST.



(a) Cover Type (b) Diabetes (c) Marketing Campaign

(d) Ship Performance (e) MNIST

Fig. 3: Itakura-Saito Divergence loss using estimated LID is used to replace KL divergence loss

Results

Learning cluster-friendly latent space

- These results are produced by a model that uses both the IS loss and KL loss along with reconstruction and K-means losses.
- For Cover Type dataset, while clusters were well-defined, the IS loss overly tightened them
- Performance declined on diabetes dataset
- MNIST performance improved, while Fashion-MNIST performed better than baseline

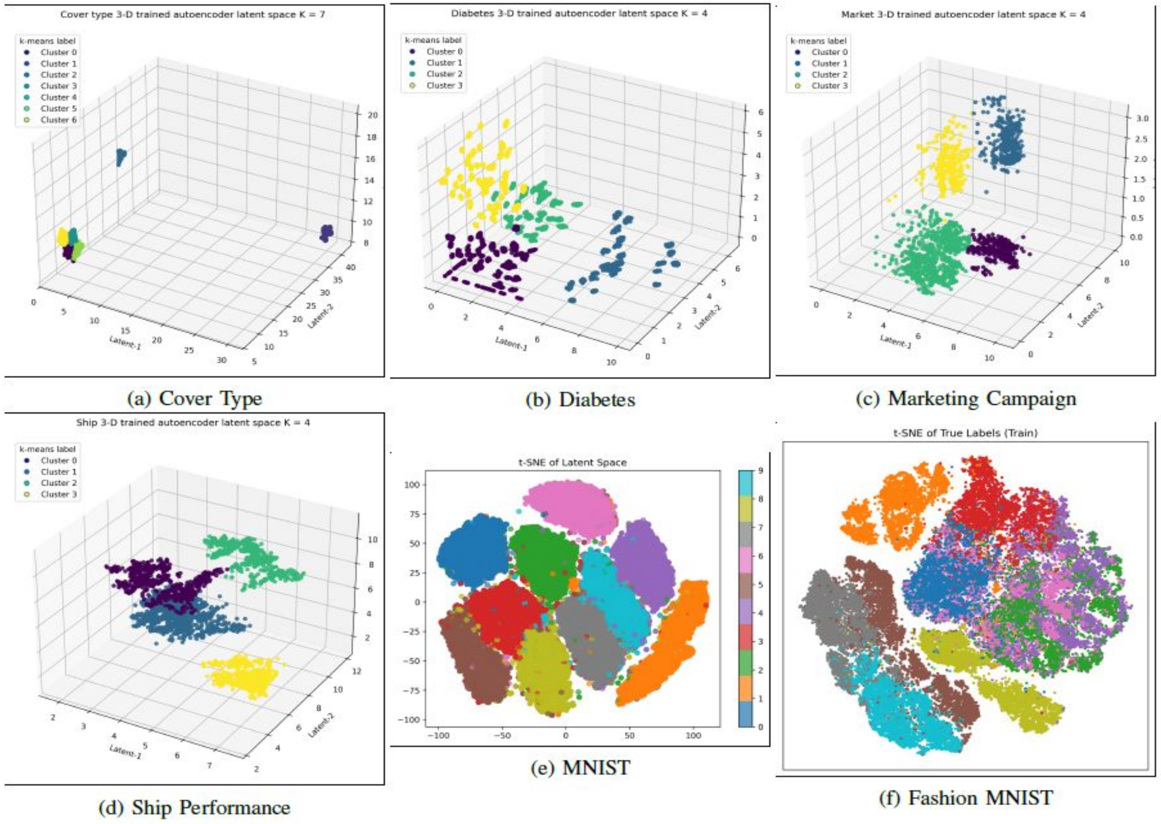


Fig. 4: Itakura-Saito Divergence loss using estimated LID used with KL divergence loss

Limitations and future work

- **LID ≠ TC as a disentanglement metric:** Using local intrinsic dimensionality as a stand-in for Total Correlation proved both theoretically unsound and empirically weak; LID still hints at entanglement (local ID > 1) but needs alternative, radius-stable formulations to be useful.
- **Multi-modal disentanglement:** Independent encoders/decoders can map each modality to a shared latent space, letting models align visual regions with text (and similar cross-modal links) for better interpretability and reasoning.
- **Limits of LID estimation:** In small (number of samples) datasets or high-dimensional datasets, Two-NN and MLE LID estimates become numerically unstable; a single global ID is inadequate for data on multiple or nested manifolds, so LID should not be optimized as a loss term in these cases.
- **Post-hoc cluster refinement:** The techniques studied adjust cluster geometry after labels are set, a strategy transferable to other algorithms to yield tighter, smoother, more separable clusters without re-assigning membership.

Conclusion

This report explored the theoretical foundations and empirical effectiveness of utilizing Local Intrinsic Dimensionality (LID) to achieve disentangled and cluster-friendly latent representations in autoencoders. The results indicate that directly substituting Total Correlation with LID does not provide theoretically sound or empirically effective disentanglement.

Nevertheless, incorporating Itakura–Saito Divergence alongside Kullback–Leibler Divergence has shown promising results in learning latent spaces conducive to clustering, notably improving cluster compactness and separability.

Through comprehensive empirical analyses across diverse datasets, it was demonstrated that careful design of loss functions and architecture modifications significantly impacts the quality of latent space representations. However, challenges remain, especially in handling small datasets or those with inherently high dimensionality. Thus, future studies should focus on robust dimensionality estimation methods, multimodal data integration, and further refinement of clustering methods to fully exploit the capabilities of latent representation learning.

Resources

Images in slides 3, 5, 6 are taken from:

- https://www.ibm.com/think/topics/variational-autoencoder
- https://numxl.com/blogs/principal-component-analysis-pca-101/
- https://www.linkedin.com/pulse/understanding-clustering-algorithms-key-techniques-bill-palifka-zw8te
- https://blog.dailydoseofds.com/p/t-sne-vs-sne-whats-the-difference
- https://www.v7labs.com/blog/autoencoders-guide

Images in slides 8, 10, 11 are generated using ChatGPT o3 model.
Image in slide 7 is taken from Professor Michael Houle's lecture notes.
All other images and resources are made using the models described in the paper. The source code and datasets for the models are present in the Github repository: https://github.com/Subhodeep01/DimProj